

How Wi-Fi HaLow Will Change the Future of Industrial IoT



Robotics Efficiency



- Enhance robotic efficiency in automated warehouses and farms
- Reduce system architecture costs
- The high-grade court and lower priced equipment's flexibility, expandable, scaling and consistent performance

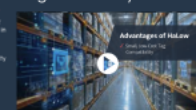
Machine Health Monitoring

- Integrate pressure and vibration sensors for predictive maintenance
- Ensure data collection integrity
- Enable analytics to predict potential machine failure for improved production line efficiency



Asset Tracking & Inventory Control

- Maximize visibility of equipment, even in challenging environments
- Enhance functionality in warehouses and cold chain logistics
- Ensure effective inventory management



SMART Factory



Robotics Efficiency



Machine Health Monitoring



Asset Tracking & Inventory Control

Presenter:
Kelly Peralta

VP of Sales & Business Development
GATEWORKS



What is going to change in the world of industrial IoT because of HaLow?

- Real Markets
- Real Applications

Industrial Markets for HaLow

HaLow technology is set to revolutionize key industrial markets by enhancing connectivity and efficiency.



SMART Transportation



SMART Energy



SMART Factories



THANK YOU



Kelly Peralta
VP of Sales & Business Development
kperalta@gateworks.com
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SMART Transportation



Rail Safety Systems



Fleet Management & Telematics

Fleet Management & Telematics

- Enable efficient monitoring of vehicle performance, location and fuel usage through advanced engine
- Allow lower costs parts using the higher efficient assets
- Enhance operational efficiency and reduce cost



Rail Safety Systems



- Connect up to 8000 nodes with 100% reliability
- Uninterruptible vehicle safety
- Enhance communication
- Increase safety, improve performance



SMART Energy



Smart Grid Monitoring



Smart Metering



Electric Vehicle Charging



Solar Inverters

Solar Inverter Applications



Electric Vehicle Charging



Smart Metering



Smart Grid Monitoring



GATEWORKS

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SMART
Transportation



SMART
Energy



SMART
Factories

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Robotics Efficiency



- Enhance robotic efficiency in automated warehouses and farms
- Reduce system architecture costs
- The high-grade court and lower-level requirements for safety, security, and power

Machine Health Monitoring

- Integrate pressure and vibration sensors for predictive maintenance
- Ensure data collection integrity
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- Allow lower costs, parts saving for higher vehicle counts
- Enhance operational efficiency and reduce cost



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SMART Energy



Smart Grid Monitoring



Smart Metering



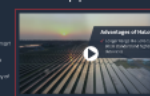
Electric Vehicle Charging



Solar Inverters

Solar Inverter Applications

- Reduce system costs
- Increase efficiency
- Enable energy monitoring
- Enable predictive maintenance



Electric Vehicle Charging

- Enable predictive maintenance
- Increase efficiency
- Enable energy monitoring
- Enable predictive maintenance



Smart Metering

- Higher number of nodes
- Enable energy monitoring
- Enable predictive maintenance
- Enable predictive maintenance



Smart Grid Monitoring

- Enable predictive maintenance
- Increase efficiency
- Enable energy monitoring
- Enable predictive maintenance



GATEWORKS

SMART Factory



Robotics
Efficiency

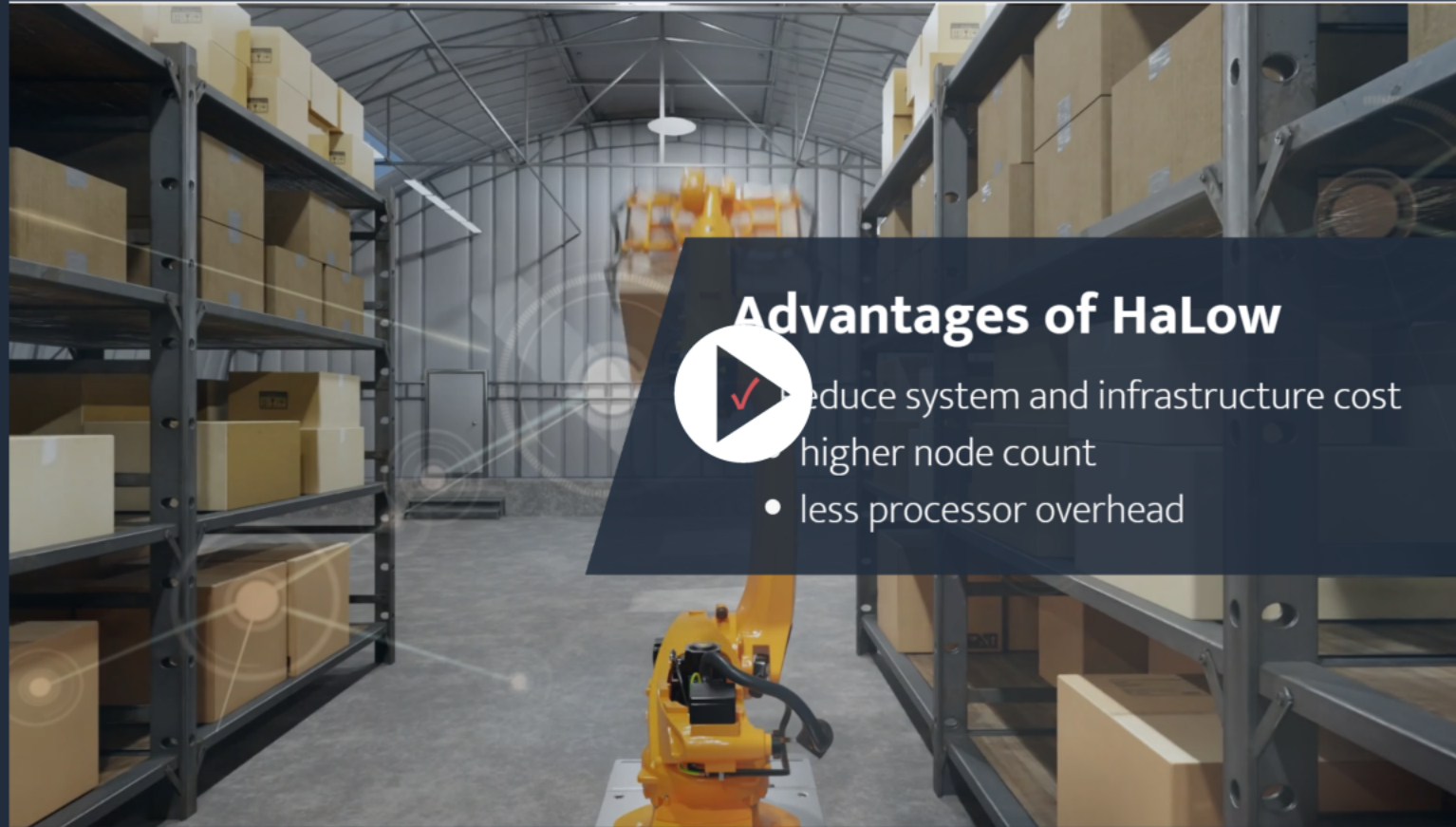


Machine Health
Monitoring



Asset Tracking &
Inventory Control

Robotics Efficiency



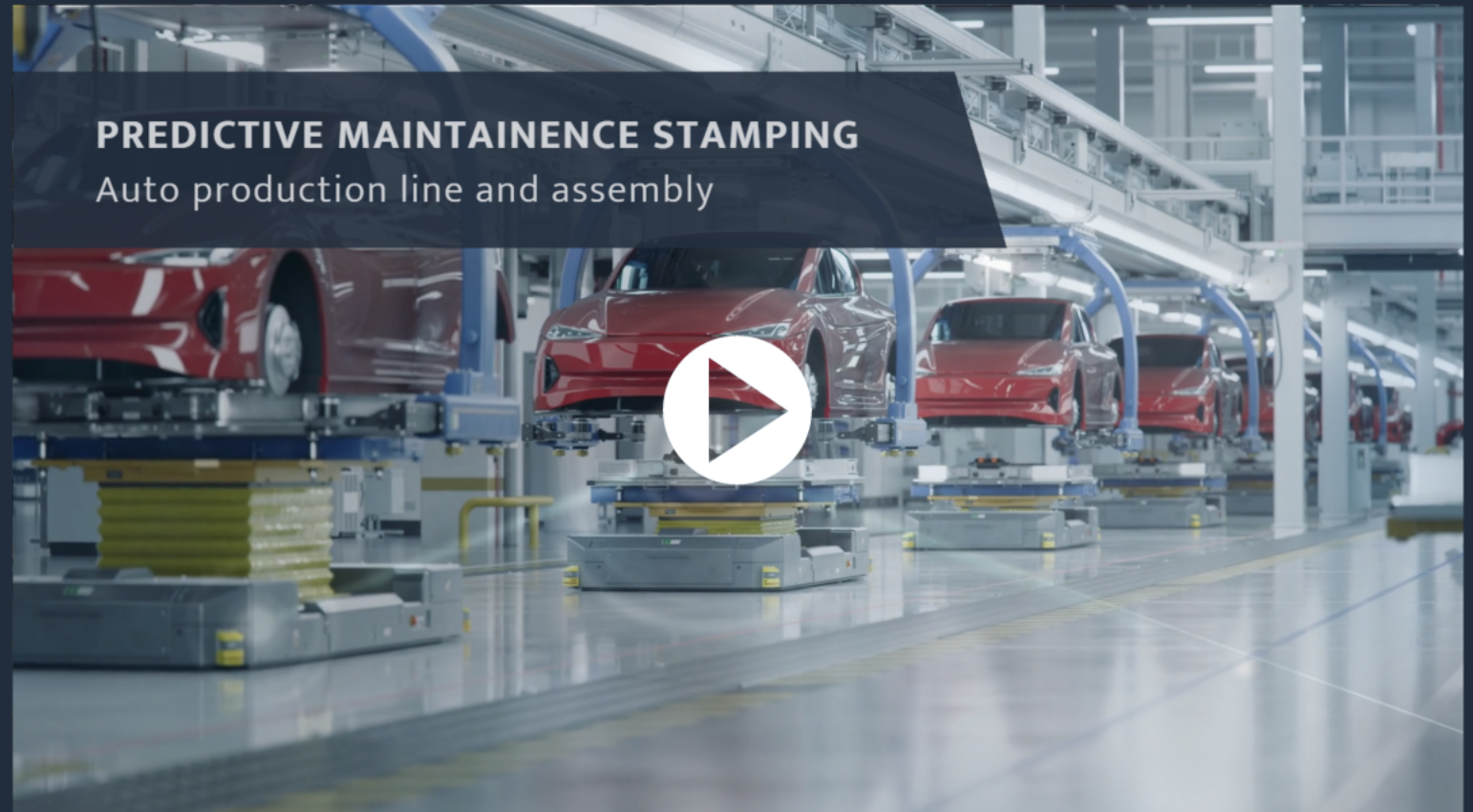
Advantages of HaLow

- Reduce system and infrastructure cost
- higher node count
- less processor overhead

- Enhance robotics efficiency in automated warehouses and AMRs
- Reduce system infrastructure costs
- The high node count and lower power requirements facilitate streamlined pallet picking and movement within factories.

Machine Health Monitoring

- Integrate pressure and vibration sensors for preemptive maintenance
- Ensure data collection integrity
- Enable analytics to predict potential machine failures and optimize production line efficiency



Asset Tracking & Inventory Control

- Maximize visibility of critical assets, even in challenging environments
- Enhance functionality in warehouses and cold chain logistics
- Ensure effective inventory management



Advantages of HaLow

- ✓ Small, Low-Cost Tag Compatibility

SMART Energy



Smart Grid
Monitoring



Smart Metering



Electric Vehicle
Charging



Solar Inverters

Smart Grid Monitoring

SMART GRID MONITORING

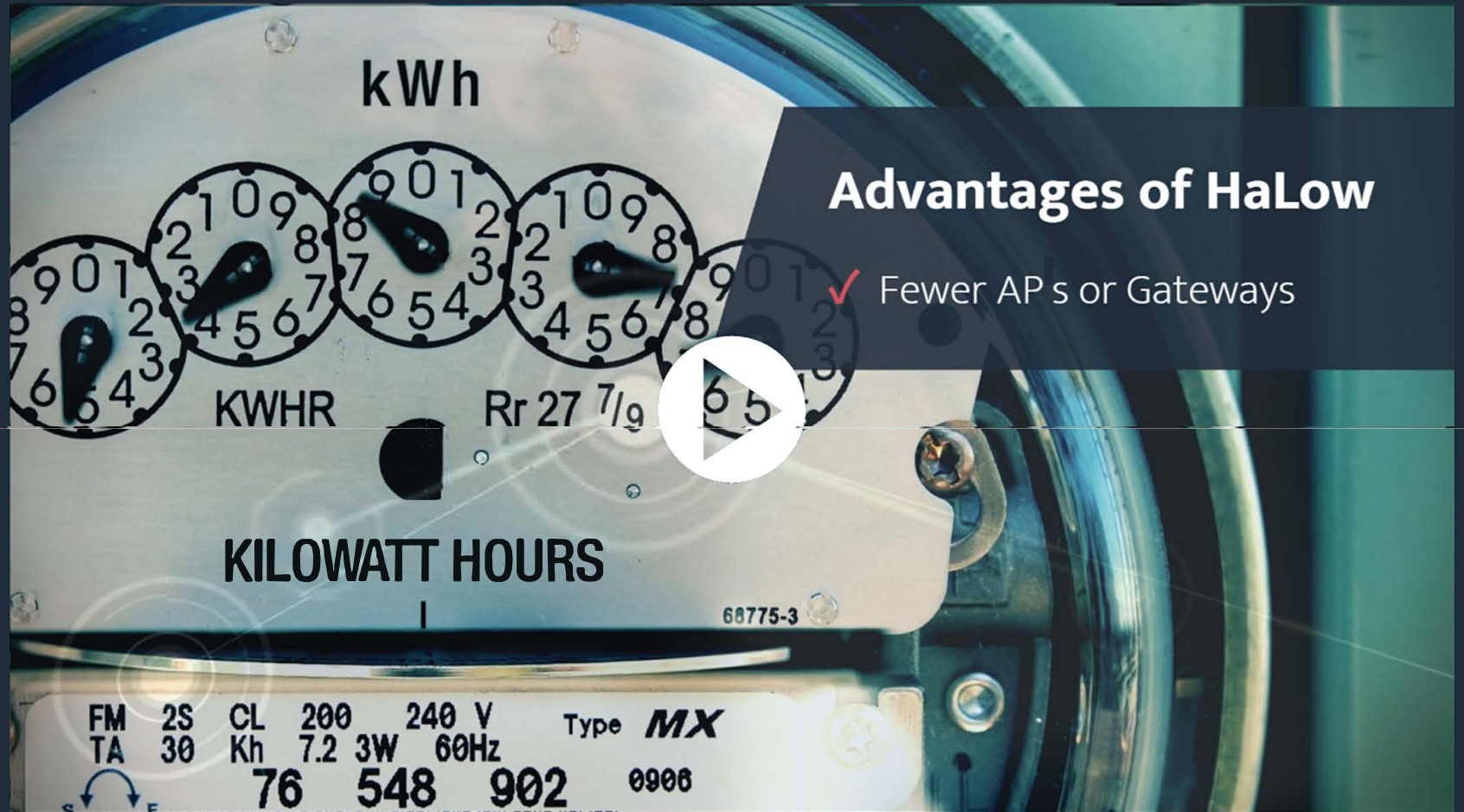
Monitor grid parameters like line sag, temperature and electrical load.



- Enable real-time monitoring of grid parameters such as line sag, temperature and electrical load
- Enhance operational efficiency
- With higher throughput, detailed sensor data can be transmitted without relying on cellular connectivity
- Optimize grid management and fault prevention

Smart Metering

- Higher number of meters connected with fewer access points
- Efficient data collection
- Enable security monitoring
- Reduce dependence on costly cellular services



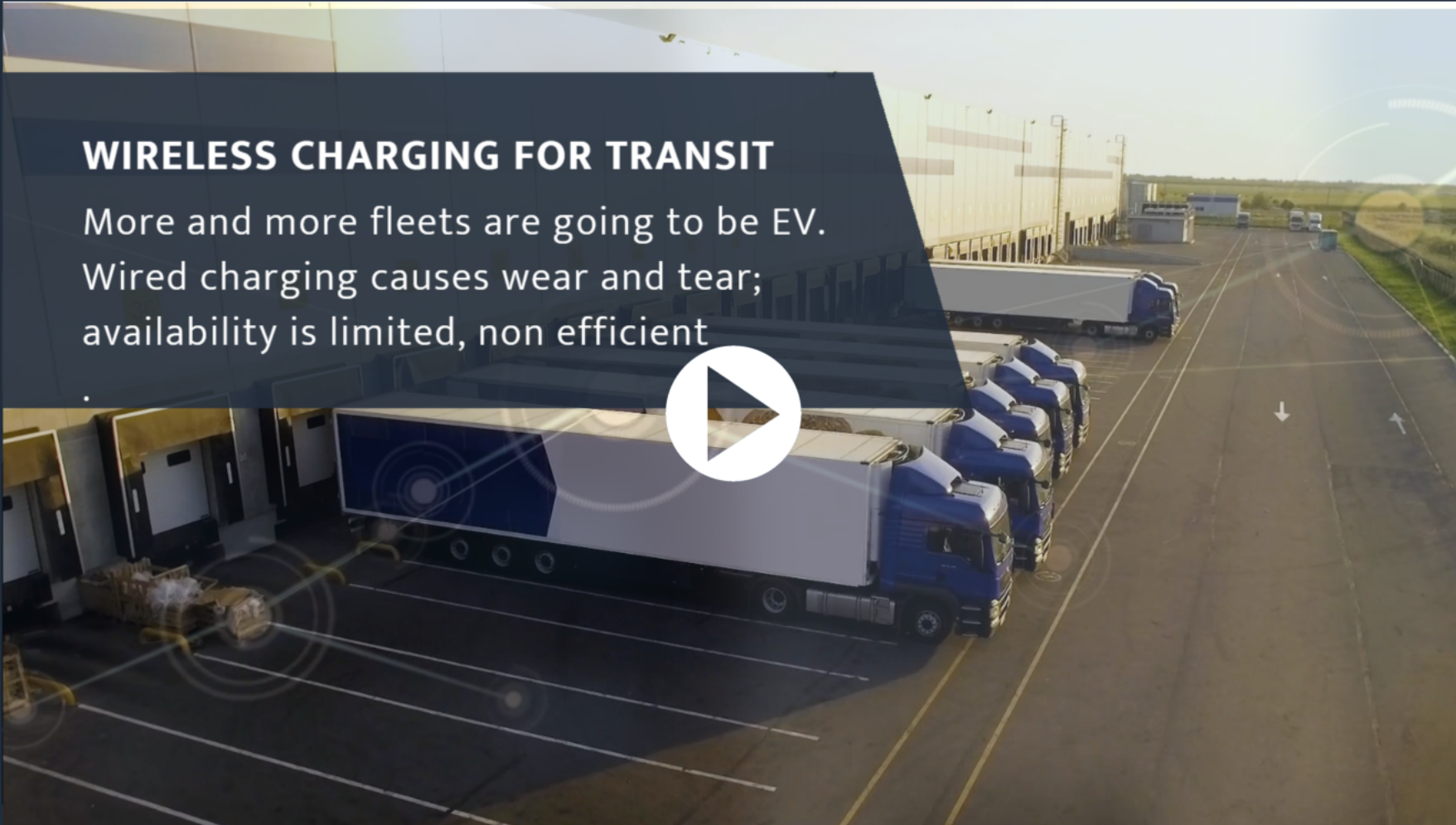
Electric Vehicle Charging

WIRELESS CHARGING FOR TRANSIT

More and more fleets are going to be EV.
Wired charging causes wear and tear;
availability is limited, non efficient



- Address limitations of wired options
- Leverage existing Wi-Fi infrastructure
- Enhance safety, reduces wear on vehicles
- Provide efficient solutions for monitoring vehicle locations and availability at long distances



Solar Inverter Applications

- Robust data transmission for performance monitoring
- Integrate within smart energy solutions
- Promote real-time analysis
- Increase efficiency of solar energy utilization



Advantages of HaLow

- ✓ Longer Range like LoRa but 802.11 standard and higher data rates

SMART Transportation



Rail Safety
Systems



Fleet Management
& Telematics

Rail Safety Systems

MAINTENANCE OF WAY SAFETY

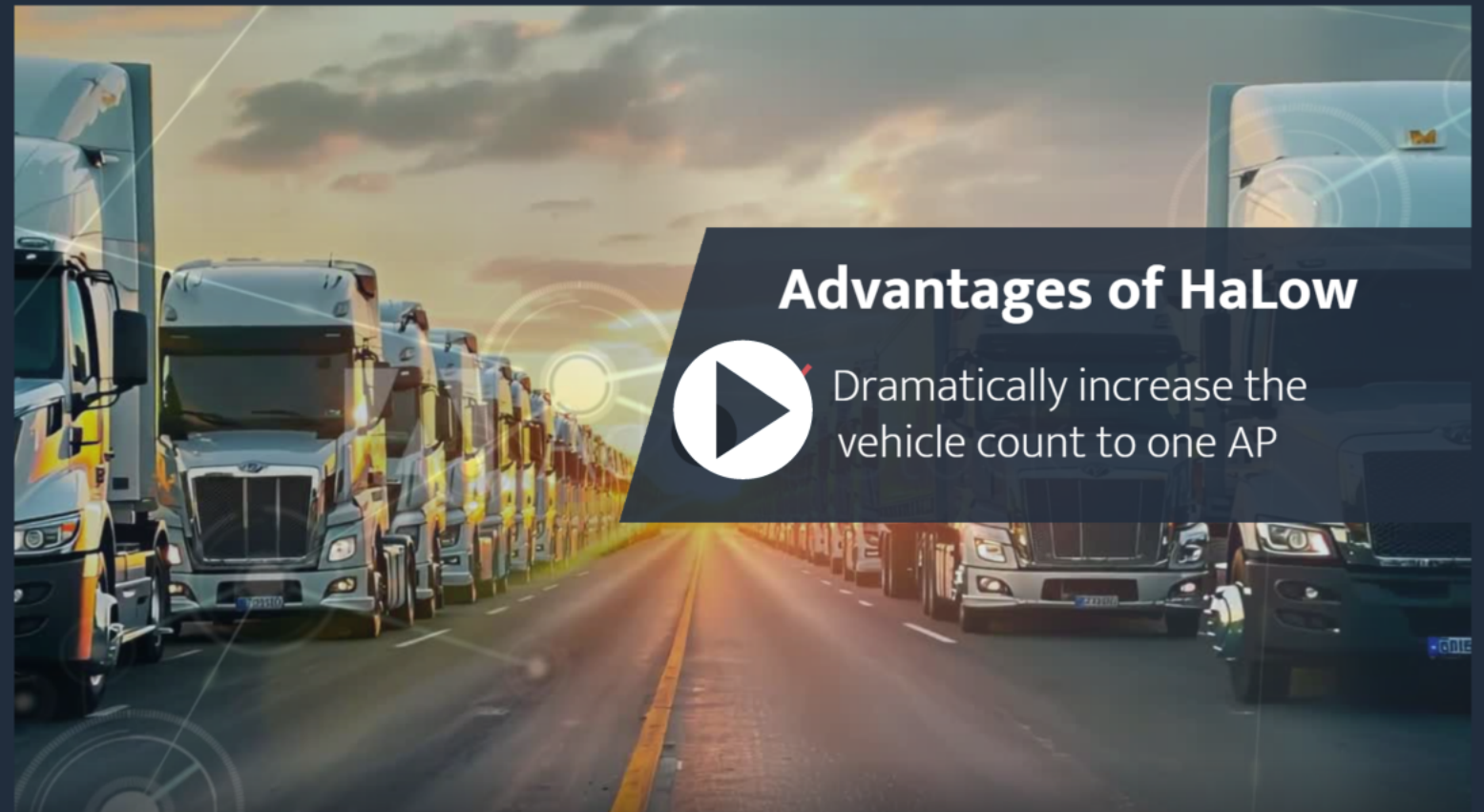
Monitor the front and rear of each vehicle or train with sensors and cameras.



- Connect up to 8000 nodes using existing wireless infrastructure
- Drastically reduce latency
- Enhance bandwidth
- Ensure timely response to potential hazards

Fleet Management & Telematics

- Enable efficient monitoring of vehicle maintenance, location and fuel usage through extended range
- Utilize fewer access points (APs) for higher vehicle counts
- Enhance operational efficiency and reduce cost



NXP

Gold
Partner



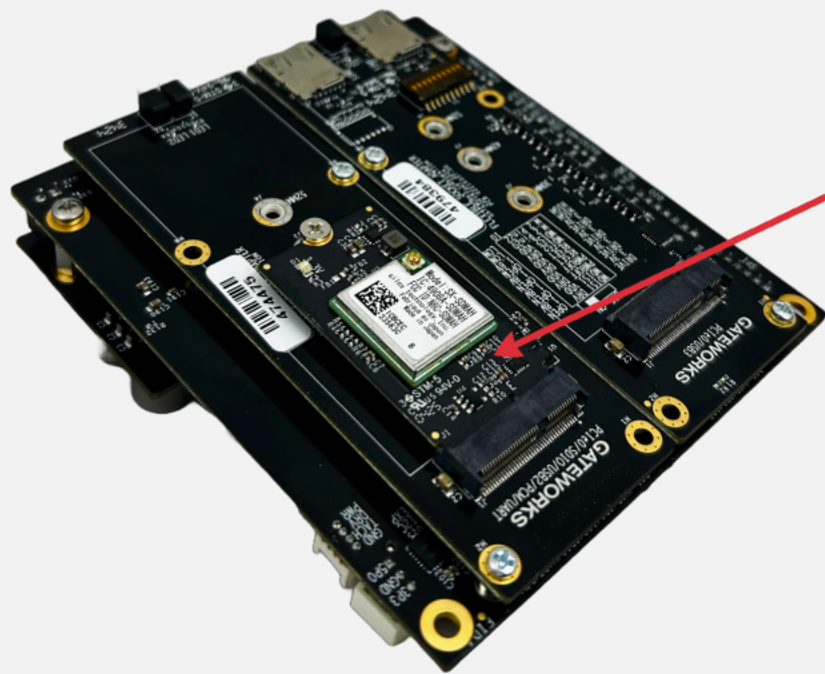
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INNOVATIONS FOR EMBEDDED DESIGN

NXP ARM INDUSTRIAL SINGLE BOARD COMPUTERS



GET 25% OFF Industrial-Grade Plug & Play Wi-Fi HaLow Development Kit



Comes with a
Wi-Fi HaLow M.2
Card powered
by the Morse
Micro MM6108
System-on-Chip



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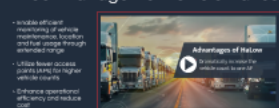


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